



ROOT



MATLAB Geek

Summer Bootcamp | 2026

WiseDuck



Code is poetry written for
machines!

Presented by



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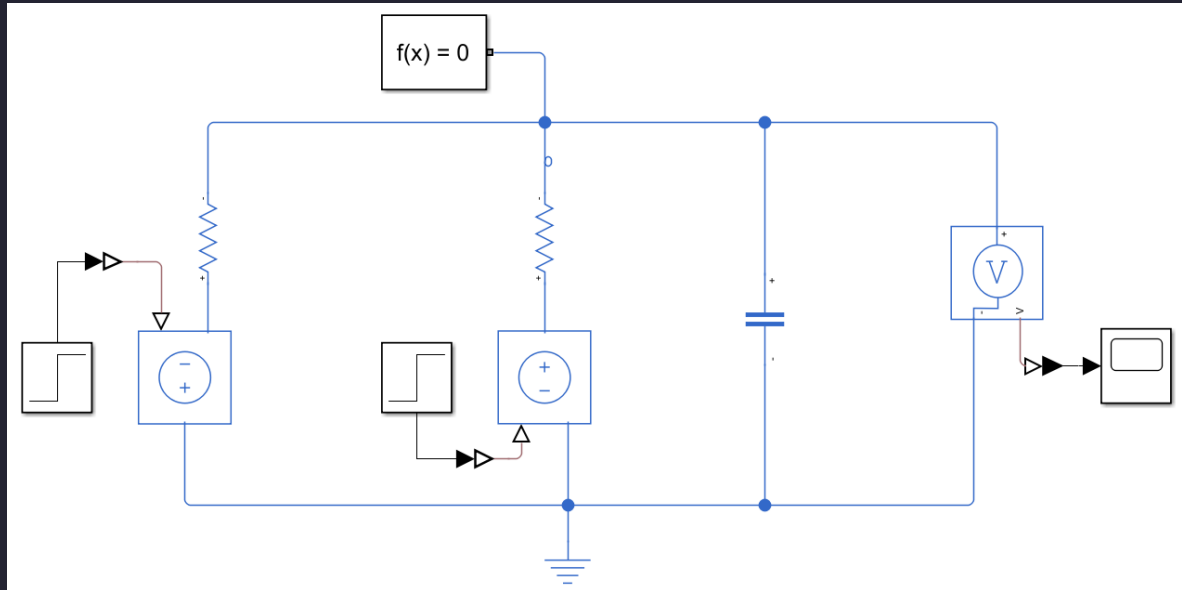
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Simulink Geek Workshop



EXAMPLE 5,0 | RC Circuit with Step Input





EXAMPLE 5,1 | Series RLC

$$a \ddot{I} + b \dot{I} + c I = 0$$

$$\alpha = b/2a = \frac{R}{2L}$$

$$\omega_0 = \sqrt{c/a} = \frac{1}{\sqrt{LC}}$$

$$at^2 + bt + c = 0$$

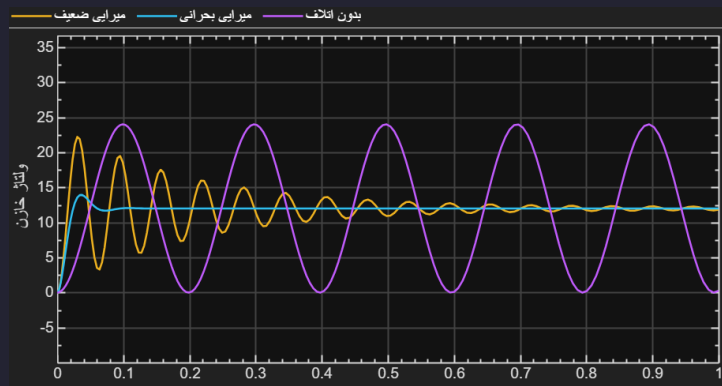
$$t = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = 0 \rightarrow \text{Undamped} \rightarrow I(t) = A_1 \cos(\omega_0 t) + A_2 \sin(\omega_0 t)$$

$$\alpha < \omega_0 \rightarrow \text{Underdamped} \rightarrow I(t) = e^{-\alpha t} (A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t))$$

$$\alpha = \omega_0 \rightarrow \text{Critically Damped} \rightarrow I(t) = (A_1 + A_2 t) e^{-\alpha t}$$

$$\alpha > \omega_0 \rightarrow \text{Overdamped} \rightarrow I(t) = A_1 e^{t_1 t} + A_2 e^{t_2 t}$$





EXAMPLE 5,2 | Series RLC – Phasor Analysis

Given:

$$R = 1\Omega, L = 1H, C = 0.02533F$$

$$V_{peak} = 12V, f = 1Hz$$

Angular frequency:

$$\omega = 2\pi f = 6.283 \text{ rad/s}$$

Reactances:

$$X_L = \omega L = 6.283\Omega$$

$$X_C = \frac{1}{\omega C} = 6.283\Omega$$

$$X_L = X_C \Rightarrow \text{Resonance}$$

Total impedance:

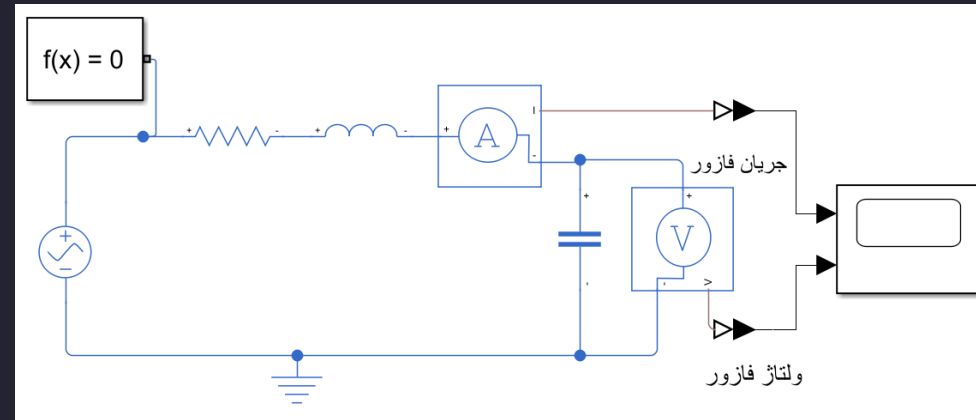
$$Z = R + j(X_L - X_C) = \boxed{1\Omega}$$

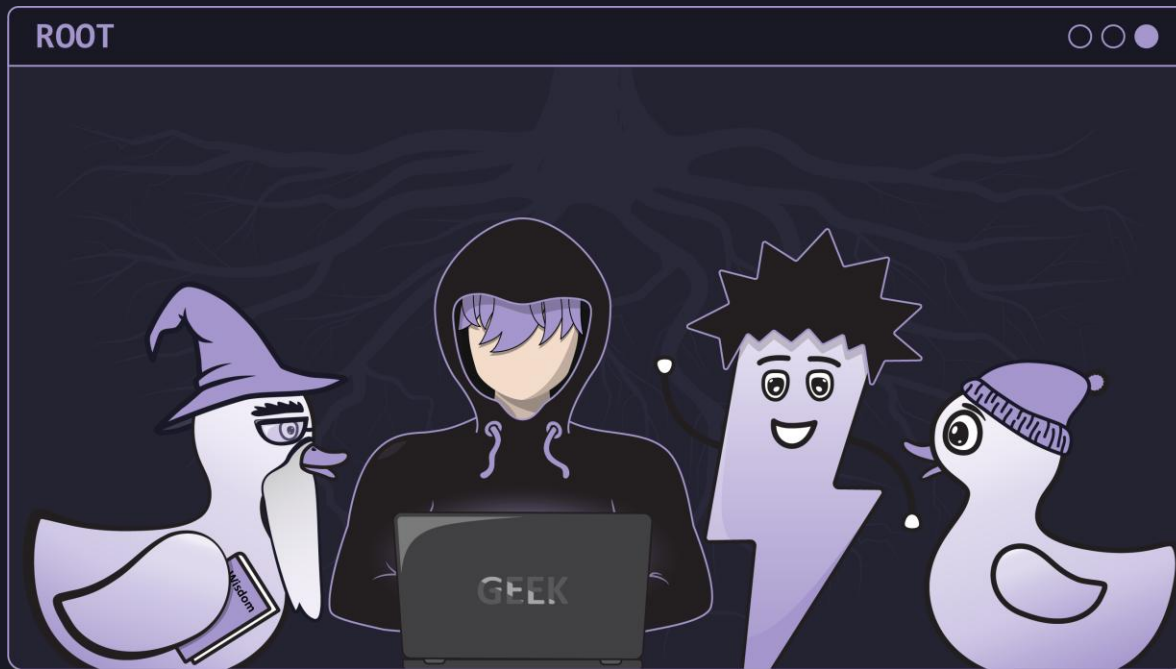
Current:

$$I_{peak} = \frac{12}{1} = 12A$$

Capacitor voltage:

$$\begin{aligned} V_{C,peak} &= I_{peak} X_C \\ &= 12(6.283) = \boxed{75.40V} \end{aligned}$$







The End!



THANKS!

Do you have any questions?



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